

RETAIL SALES & INVENTORY ANALYSIS

Full Project Report

Excel | MySQL | Power BI

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Phase 1: Excel Data Cleaning

Nine raw files were loaded into Excel, inspected for issues, cleaned, and saved as CSV files for SQL import.

1.1 Nine Tables — Raw Data Overview

Table	Rows	Issues Found
customers	1,445	Phone 87.7% NULL, zip_code as integer, mixed casing
products	321	brand_id and category_id are numbers only
orders	1,615	170 NULL shipped_dates, order_status as 1-4 only
order_items	4,722	Discount stored as "20%" text, not decimal
staffs	10	Manager_id NULL for top manager, stored as float
stores	3	zip_code as integer, no region column
brands	9	Clean — no issues found
categories	7	Clean — no issues found
stocks	939	Product name missing — ID only

1.2 Cleaning Steps

Step 1 — Remove Empty Columns

- The customers file had 7 empty unnamed columns leftover from the source system
- Select columns J to P, right-click, then click Delete

Step 2 — Fix NULL Values

Table	Column	Nulls	Fix Applied
customers	phone	1,267	Replaced NULL with N/A text
orders	shipped_date	170	Filled with Not Shipped for pending orders
staffs	manager_id	1	Filled with 0 — top manager has no manager

Step 3 — Fix Data Types

Column	Table	Wrong Type	Correct Type
zip_code	customers, stores	INT	VARCHAR(10) — text to keep leading zeros
list_price	order_items	TEXT	DECIMAL(10,2) — formatted as currency
discount	order_items	"20%" text	Fixed with REPLACE() in SQL queries
shipped_date	orders	NULL	Text Not Shipped for unshipped orders

Step 4 — Normalize Text Fields

Field	Issue	Excel Formula
first_name, last_name, city	Mixed casing	=PROPER(TRIM(A2))
state	Inconsistent case	=UPPER(TRIM(G2))
email	Mixed uppercase	=LOWER(TRIM(E2))
zip_code	Stored as integer	=TEXT(I2,"00000")

product_name	Extra spaces	=TRIM(B2)
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Step 5 — Check for Duplicates

- Duplicate rows: 0 across all 9 tables
- Duplicate order_id: 0 in orders table
- Duplicate product_id: 0 in products table
- 30 duplicate product names found — same product in multiple categories, intentional

Step 6 — Save as CSV for SQL Import

- File, Save As, then CSV Comma delimited for each of the 9 files
- Import order: stores, brands, categories, staffs, customers, products, orders, order_items, stocks

Phase 2: SQL Database and Queries

2.1 Database Setup

Commands run in MySQL Workbench:

```
CREATE DATABASE retail_analysis;
```

```
USE retail_analysis;
```

All 9 tables imported using Table Data Import Wizard. Tables renamed to short names:

```
RENAME TABLE stores-retail sales and inventory analysis TO stores;
```

Repeat this command for all 9 tables.

2.2 ER Diagram — 11 Relationships

Child Table	FK Column	Parent Table	Meaning
products	brand_id	brands	Many products belong to one brand
products	category_id	categories	Many products in one category
staffs	store_id	stores	Many staff work at one store
staffs	manager_id	staffs	Self-reference: staff has one manager
orders	customer_id	customers	Many orders placed by one customer
orders	store_id	stores	Many orders placed at one store
orders	staff_id	staffs	Many orders handled by one staff
order_items	order_id	orders	Many items belong to one order
order_items	product_id	products	Many items reference one product
stocks	store_id	stores	Many stock records at one store
stocks	product_id	products	Many stock records for one product

2.3 Key SQL Queries

Query 1 — Total Sales

```
SELECT COUNT(DISTINCT o.Order_id) AS total_orders,  
       SUM(oi.Quantity) AS total_items_sold,  
       ROUND(SUM(oi.List_price * oi.Quantity  
               * (1 - REPLACE(oi.Discount,'%','')/100)), 2) AS total_revenue  
FROM orders o  
JOIN order_items oi ON o.Order_id = oi.Order_Id  
WHERE o.Order_status = 4;
```

total_orders	total_items_sold	total_revenue	avg_order_value
1,445	6,318	\$6,662,615.24	\$1,581.07

Query 2 — Sales by Store

```
SELECT s.store_name, s.city, s.state,  
       COUNT(DISTINCT o.Order_id) AS total_orders,  
       ROUND(SUM(oi.List_price * oi.Quantity
```

```

* (1 - REPLACE(oi.Discount,'%')/100)), 2) AS net_revenue
FROM stores s
JOIN orders o ON s.store_id = o.Store_id
JOIN order_items oi ON o.Order_id = oi.Order_id
WHERE o.Order_status = 4
GROUP BY s.store_name, s.city, s.state
ORDER BY net_revenue DESC;

```

Store	City	State	Orders	Net Revenue
Baldwin Bikes	Baldwin	NY	1,019	~\$4,500,000
Santa Cruz Bikes	Santa Cruz	CA	284	~\$1,300,000
Rowlett Bikes	Rowlett	TX	142	~\$862,000

Query 3 — Top 10 Products

```

SELECT p.Product_name, b.Brand_name, c.Category_name,
SUM(oi.Quantity) AS qty_sold,
ROUND(SUM(oi.List_price * oi.Quantity
* (1 - REPLACE(oi.Discount,'%')/100)), 2) AS revenue
FROM order_items oi
JOIN orders o ON oi.Order_id = o.Order_id
JOIN products p ON oi.Product_id = p.Product_id
JOIN brands b ON p.Brand_id = b.Brand_id
JOIN categories c ON p.Category_id = c.Category_id
WHERE o.Order_status = 4
GROUP BY p.Product_name, b.Brand_name, c.Category_name
ORDER BY revenue DESC LIMIT 10;

```

Query 4 — Top 10 Highest Spending Customers

```

SELECT CONCAT(c.First_name,' ',c.Last_name) AS customer_name,
c.City, c.State,
ROUND(SUM(oi.List_price * oi.Quantity
* (1 - REPLACE(oi.Discount,'%')/100)), 2) AS total_spent
FROM customers c
JOIN orders o ON c.Customer_id = o.Customer_id
JOIN order_items oi ON o.Order_id = oi.Order_id
WHERE o.Order_status = 4
GROUP BY c.Customer_id, customer_name, c.City, c.State
ORDER BY total_spent DESC LIMIT 10;

```

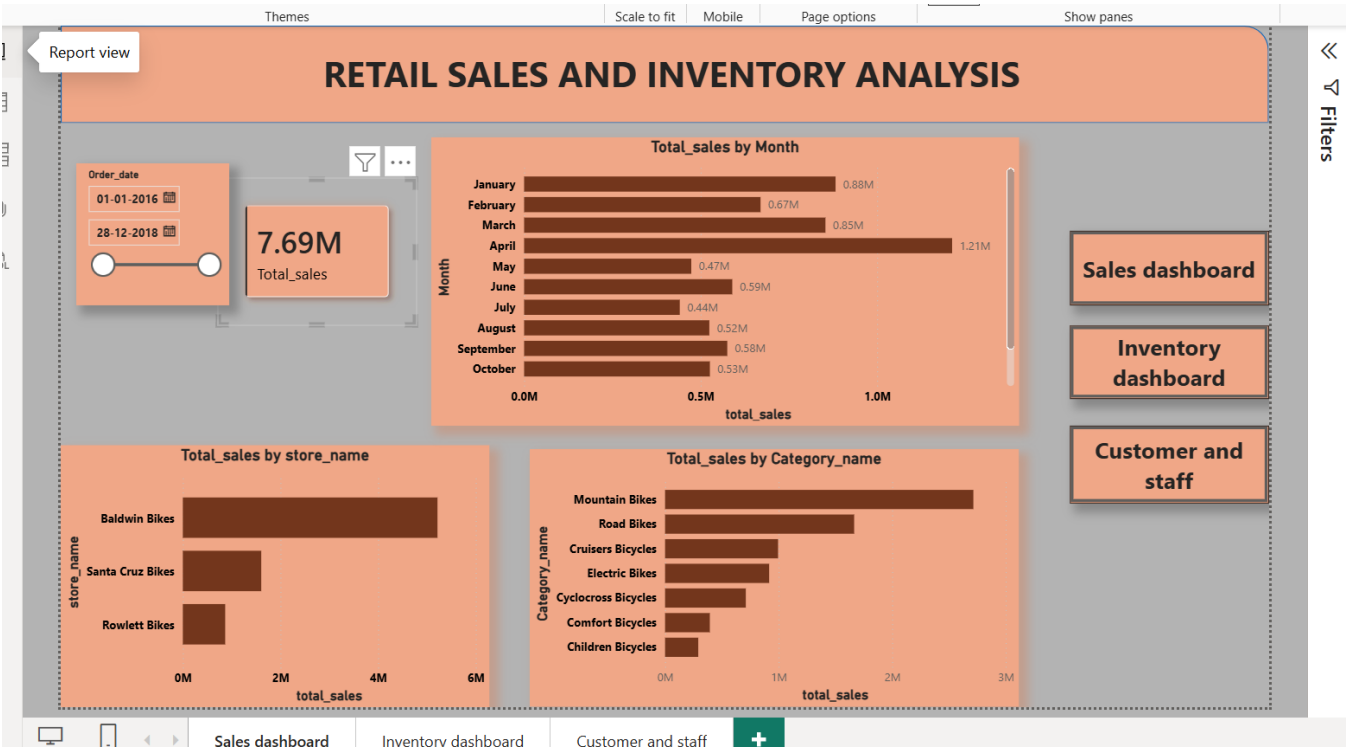
Phase 3: Power BI Dashboard

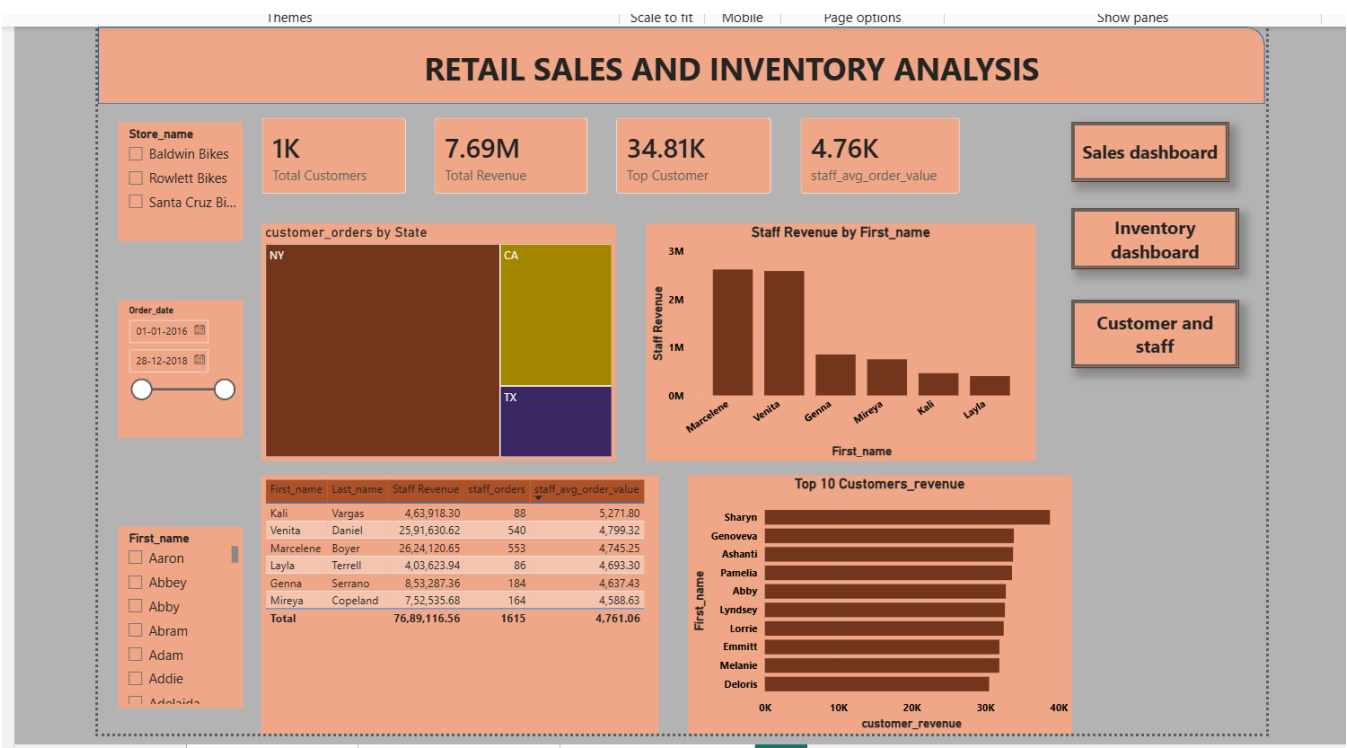
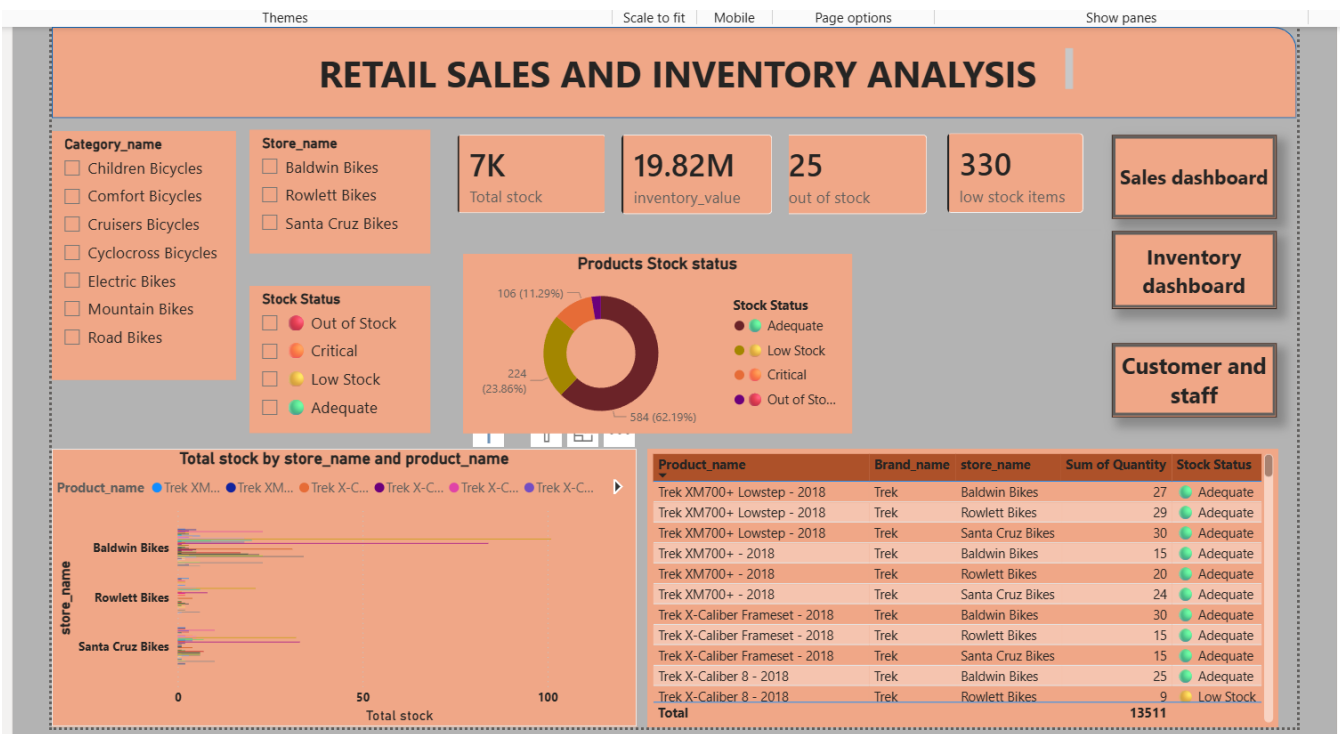
3.1 Connect Power BI to MySQL

Step	Action	Detail
1	Open Power BI Desktop	File then New Report
2	Get Data	Home, Get Data, then MySQL Database
3	Enter connection	Server: localhost Database: retail_analysis
4	Login	Username: root Password: your password
5	Select tables	Tick all 9 tables then click Load
6	Set relationships	Model view, drag FK column to PK column between tables

3.2 Dashboard Pages

Page	Title	Key Visuals
1	Executive Summary	Revenue KPI, Orders KPI, Avg Order Value card, Discount % card
2	Sales by Store	Bar chart by store, Map by state, Revenue share donut chart
3	Product Performance	Top 10 products table, Category comparison bar, Brand chart
4	Customer Analysis	Spending tier breakdown, Top 10 spenders table by revenue
5	Order Analysis	Status donut chart, Monthly trend line, Late shipments bar
6	Stock and Inventory	Low stock alert table, Stock quantity by store bar chart





3.3 Key DAX Measures

Net Revenue =

SUMX(order_items, order_items[List_price] * order_items[Quantity]
 * (1 - VALUE(SUBSTITUTE(order_items[Discount], "%", "")) / 100))

Avg Order Value = DIVIDE([Net Revenue], DISTINCTCOUNT(orders[Order_id]))

Late Shipment % = DIVIDE(
COUNTROWS(FILTER(orders, orders[Shipped_date] > orders[Required_date])),
COUNTROWS(orders)) * 100

Business Insights

4.1 Store Performance

Store	Region	Orders	Revenue Share	Late Rate	Rating
Baldwin Bikes	East NY	1,019	67.6%	31.2%	TOP PERFORMER
Santa Cruz Bikes	West CA	284	19.5%	36.6%	AVERAGE
Rowlett Bikes	South TX	142	12.9%	26.0%	NEEDS ATTENTION

- Baldwin Bikes NY drives 67.6% of all revenue from a single store
- Santa Cruz has the worst late shipment rate at 36.6% — needs urgent attention
- Rowlett TX is underperforming with only 8.8% of orders despite large state market

4.2 Top Selling Products

- Trek dominates — 42% of catalog, prices ranging \$379 to \$11,999
- Road Bikes and Mountain Bikes are the top revenue categories
- 2018 model year products outsell all others — 204 of 321 total products
- Budget products under \$500 make up 33% of catalog but lower revenue per unit

4.3 Products Needing Restocking

Priority	Criteria	Action Required
CRITICAL	Quantity = 0 in any store	Reorder immediately — active lost sales risk
HIGH	Quantity 5 or less for top sellers	Reorder within 7 days
MEDIUM	Quantity 10 or less	Reorder within 30 days
LOW	Quantity above 10	Monitor only — no action needed now

SQL query to find low stock products:

```
SELECT p.Product_name, s.Quantity, st.store_name
FROM stocks s
JOIN products p ON s.Product_id = p.Product_id
JOIN stores st ON s.Store_id = st.store_id
WHERE s.Quantity <= 5 ORDER BY s.Quantity ASC;
```

4.4 Staff Performance

- Staff 6 and Staff 7 at Baldwin NY handle approximately 68% of all orders combined
- Overall late shipment rate of 31.7% suggests a warehouse process issue not staff issue
- Rowlett TX staff handle the fewest orders — capacity available for expansion

Business Recommendations

5.1 Increase Stock for High-Demand Products

#	Action	Priority
1	Restock Trek Road Bikes at Baldwin NY — highest revenue risk if out of stock	CRITICAL
2	Set auto-reorder trigger when stock drops below 5 units for premium products	HIGH
3	Prioritize 2018 model inventory — outsells all other model years	HIGH
4	Share excess inventory across stores — Baldwin can transfer stock to TX and CA	MEDIUM

5.2 Improve Low-Performing Stores

#	Store	Recommendation	Expected Result
1	Rowlett TX	Run targeted Texas marketing campaigns. Large state, low 8.8% penetration.	+25% orders
2	Santa Cruz CA	Add staff, improve local digital marketing, run seasonal promotions.	+15% orders
3	All stores	Cross-store inventory sharing to prevent stockouts in any location.	Less lost sales

5.3 Reduce Shipment Delays

#	Issue	Recommendation	Target
1	31.7% late rate overall	Audit full fulfillment process — packing, dispatch, carrier.	Below 10%
2	Santa Cruz worst at 36.6%	Review warehouse staffing and processes at Santa Cruz only.	Below 15%
3	No daily SLA tracking	Add daily shipping dashboard for store managers.	Daily review
4	Tight required dates set	Add 1-day buffer when entering required date at order time.	Quick win

5.4 Promotions for Slow-Moving Products

#	Product Type	Promotion Strategy	Discount
1	2016 model year 26 products	Bundle with accessories or clearance to free shelf space.	30 to 40%
2	Children Bicycles	Back-to-school campaign July to September with family deal.	15%
3	Luxury bikes above \$5K	Target cycling clubs and corporate buyers with B2B pricing.	B2B deals
4	Electric Bikes	Offer EMI payment options and free test-ride events.	EMI options
5	Strider and Heller brands	Bundle with popular Trek purchases as gift-with-purchase.	Bundle deals

Summary and Next Steps

Key Finding	Value
Total Net Revenue (completed orders)	\$6,662,615
Total Orders Completed	1,445 of 1,615 placed
Average Order Value	\$1,581.07
Average Discount Given	10.6% per order
Top Store	Baldwin Bikes NY — 67.6% of revenue
Late Shipment Rate	31.7% — critical issue to fix
Repeat Customers	0 — all customers are one-time buyers
Top Brand	Trek — 42% of catalog
Top Model Year	2018 — 204 of 321 products

Priority Action List

Priority	Action	Owner
1 — URGENT	Fix 31.7% late shipment rate — audit Santa Cruz process	Operations
2 — URGENT	Launch customer loyalty program — zero repeat buyers currently	Marketing
3 — HIGH	Expand Rowlett TX marketing — large market, very low penetration	Marketing
4 — HIGH	Restock Trek Road Bikes at Baldwin before stockout occurs	Inventory
5 — MEDIUM	Clear 2016 model inventory with 30 to 40% promotional discount	Sales
6 — MEDIUM	Set auto-reorder triggers for top 50 highest revenue products	Inventory
7 — ONGOING	Share Power BI dashboard with all store managers monthly	Analytics